# Independence

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**Definition 0.1** (Statistical independence)**.** Random variables $X_1, \dots, X_n$ are **statistically independent** [1] if, for all sets of real numbers $A_1, \dots, A_n$, the probability that every $X_i$ falls in its set is the product of the individual probabilities:

$$\Pr(X_1 \in A_1, \dots, X_n \in A_n) = \prod_{i=1}^{n} \Pr(X_i \in A_i)$$

Remark

*Remark.* We write $X \perp\!\!\!\perp Y$ for "$X$ and $Y$ are independent". The symbol $\perp\!\!\!\perp$ is essentially $\prod$ upside-down, which can remind you of the definition.

**Theorem 0.1** (Independence of discrete random variables: the joint PMF factors)**.** *Discrete random variables $X_1, \dots, X_n$ are statistically independent if and only if their joint PMF factors into the product of their PMFs: for all real numbers $x_1, \dots, x_n$,*

$$\mathbb{P}(X_1 = x_1, \dots, X_n = x_n) = \prod_{i=1}^{n} \mathbb{P}(X_i = x_i)$$

Proof

*Proof.* **Only if.** Take $A_i = \{x_i\}$ for each $i$ in Definition 0.1.
**If.** Let $A_1, \dots, A_n$ be sets of real numbers, and let $C$ be the set of tuples $(x_1, \dots, x_n)$ with each $x_i \in A_i \cap \mathcal{R}(X_i)$. Each range $\mathcal{R}(X_i)$ is countable, so $C$ is countable. Each $X_i$ takes its values in $\mathcal{R}(X_i)$, so the event $\{X_1 \in A_1, \dots, X_n \in A_n\}$ is the disjoint union of the events $\{X_1 = x_1, \dots, X_n = x_n\}$ over $(x_1, \dots, x_n) \in C$:

$$
\begin{aligned}
\Pr(X_1 \in A_1, \dots, X_n \in A_n) &= \sum_{(x_1,\dots,x_n)\in C} \mathbb{P}(X_1 = x_1, \dots, X_n = x_n) && \text{(countable additivity over the disjoint even} \\
&= \sum_{(x_1,\dots,x_n)\in C} \prod_{i=1}^{n} \mathbb{P}(X_i = x_i) && \text{(the joint PMF factors)} \\
&= \prod_{i=1}^{n} \sum_{x_i \in A_i \cap \mathcal{R}(X_i)} \mathbb{P}(X_i = x_i) && \text{(a sum over the product set } C \text{ of non-nega} \\
&= \prod_{i=1}^{n} \Pr(X_i \in A_i) && \text{(countable additivity over the disjoint even}
\end{aligned}
$$

□

**Example 0.1** (Two fair coin flips)**.** Flip two fair coins, and let $X_1$ and $X_2$ indicate heads on the first and second flip. Each of the four outcomes has probability $1/4$, so for example:

$$
\begin{aligned}
\mathbb{P}(X_1 = 1, X_2 = 1) &= \tfrac{1}{4} \\
&= \tfrac{1}{2} \cdot \tfrac{1}{2} \\
&= \mathbb{P}(X_1 = 1)\, \mathbb{P}(X_2 = 1)
\end{aligned}
$$

and the same factorization holds for the other three pairs of values, so $X_1 \perp\!\!\!\perp X_2$ by Theorem 0.1. In contrast, $X_1$ and the total number of heads, $X_1 + X_2$, are not independent: $\mathbb{P}(X_1 = 0, X_1 + X_2 = 2) = 0$, but

$$
\begin{aligned}
\mathbb{P}(X_1 = 0)\, \mathbb{P}(X_1 + X_2 = 2) &= \tfrac{1}{2} \cdot \tfrac{1}{4} \\
&= \tfrac{1}{8}.
\end{aligned}
$$

**Theorem 0.2** (Independence with densities: the joint density factors)**.** ***Continuous case.*** *Continuous random variables $X$ and $Y$ with densities $\mathbb{p}(X = x)$ and $\mathbb{p}(Y = y)$ are statistically independent if and only if the product $\mathbb{p}(X = x)\, \mathbb{p}(Y = y)$ is a joint density of $X$ and $Y$.*

***Mixed case.*** *A discrete random variable $X$ and a continuous random variable $Y$ with density $\mathbb{p}(Y = y)$ are statistically independent if and only if the product $\mathbb{P}(X = x)\, \mathbb{p}(Y = y)$ is a joint density-mass function of $X$ and $Y$.*

ℹ Remark

*Remark.* The proof is beyond these notes' scope: it needs measure theory that the notes do not develop, to extend a density's integrals from intervals to general sets and to exchange the order of integration [2], [3]. The continuous case extends to $n$ random variables with a

joint density of all $n$. The statement says "a joint density" rather than "the joint density" because a density is not unique: changing it on a set of zero area (zero length, in the mixed case) changes none of its integrals.

**Example 0.2** (The PMF form fails for continuous random variables)**.** The factorization in Theorem 0.1 does not work as a definition of independence for continuous random variables: there, both sides are 0 at every point, so it would call every pair of continuous random variables independent. For instance, let $X \sim \text{Uniform}(0,1)$ (uniform distribution), whose density is 1 on $[0,1]$, and let $Y = X$. For all real numbers $x$ and $y$, the event $\{X = x,\ Y = y\}$ is $\{X = x\}$ if $y = x$ and empty otherwise, so it has probability 0, because $\Pr(X = x) = 0$ for the continuous $X$. Likewise

$$\begin{aligned}\Pr(X = x)\ \Pr(Y = y) &= 0 \cdot 0 \\ &= 0,\end{aligned}$$

so the PMF form holds. But $X$ and $Y$ are not independent:

$$\begin{aligned}\Pr(X \in [0, \tfrac{1}{2}],\ Y \in [0, \tfrac{1}{2}]) &= \Pr(X \in [0, \tfrac{1}{2}]) && (Y = X) \\ &= \int_0^{1/2} 1\, dx && (\text{the density of } X \text{ is 1 on } [0,1]) \\ &= \tfrac{1}{2} && (\text{integrate})\end{aligned}$$

while

$$\begin{aligned}\Pr(X \in [0, \tfrac{1}{2}])\ \Pr(Y \in [0, \tfrac{1}{2}]) &= \tfrac{1}{2} \cdot \tfrac{1}{2} \\ &= \tfrac{1}{4}.\end{aligned}$$

**Definition 0.2** (Conditional independence)**.** Random variables $Y_1, \ldots, Y_n$ are **conditionally independent** [4] given a discrete random variable (or vector) $\underset{\sim}{X}$ if, for every value $\underset{\sim}{x}$ with $\Pr(\underset{\sim}{X} = \underset{\sim}{x}) > 0$ and all sets of real numbers $A_1, \ldots, A_n$, the conditional probability that every $Y_i$ falls in its set is the product of the individual conditional probabilities:

$$\Pr(Y_1 \in A_1, \ldots, Y_n \in A_n \mid \underset{\sim}{X} = \underset{\sim}{x}) = \prod_{i=1}^{n} \Pr(Y_i \in A_i \mid \underset{\sim}{X} = \underset{\sim}{x})$$

When $\underset{\sim}{X}$ is continuous, every event $\{\underset{\sim}{X} = \underset{\sim}{x}\}$ has probability 0, so the condition is stated with densities instead: for every $\underset{\sim}{x}$ with $\mathbb{p}(\underset{\sim}{X} = \underset{\sim}{x}) > 0$ and all $y_1, \ldots, y_n$,

$$\frac{\mathbb{p}(\underset{\sim}{X} = \underset{\sim}{x}, Y_1 = y_1, \ldots, Y_n = y_n)}{\mathbb{p}(\underset{\sim}{X} = \underset{\sim}{x})} = \prod_{i=1}^{n} \frac{\mathbb{p}(\underset{\sim}{X} = \underset{\sim}{x}, Y_i = y_i)}{\mathbb{p}(\underset{\sim}{X} = \underset{\sim}{x})}$$

where each $\mathbb{p}(\cdot)$ is a joint density (with probability masses in place of densities for any

discrete $Y_i$, as in a joint density-mass function).

**i** Remark

*Remark.* For continuous $\underset{\sim}{X}$, each ratio in the density form is the density of the $Y$'s under their conditional distribution given $\underset{\sim}{X} = \underset{\sim}{x}$, so both forms say the same thing: given $\underset{\sim}{X} = \underset{\sim}{x}$, the $Y_i$'s joint distribution factors.

**Example 0.3** (Two tests of the same patient)**.** Let $X$ indicate whether a patient has a disease, and let $Y_1$ and $Y_2$ indicate positive results on two tests whose errors are unrelated, so that $Y_1$ and $Y_2$ are conditionally independent given $X$. Suppose $\Pr(X = 1) = 0.1$, each test is positive with probability 0.9 if $X = 1$ and with probability 0.1 if $X = 0$. Then, by the law of total probability:

$$\begin{aligned}
\Pr(Y_1 = 1, Y_2 = 1) &= (0.9)(0.9)(0.1) + (0.1)(0.1)(0.9) && \text{(condition on } X\text{; factor given } X\text{)} \\
&= 0.081 + 0.009 && \text{(multiply)} \\
&= 0.09 && \text{(add)}
\end{aligned}$$

but

$$\begin{aligned}
\Pr(Y_1 = 1) &= (0.9)(0.1) + (0.1)(0.9) \\
&= 0.18,
\end{aligned}$$

so $\Pr(Y_1 = 1)\ \Pr(Y_2 = 1) = 0.0324 \neq 0.09$: the tests are conditionally independent given $X$, but not independent.

**Example 0.4** (Two coins, given their total)**.** Let $Y_1$ and $Y_2$ indicate heads on two fair coin flips, which are independent (Example 0.1), and let $X = Y_1 + Y_2$ be the number of heads. Given $X = 1$, which has probability $1/2$, exactly one coin is heads, so:

$$\begin{aligned}
\Pr(Y_1 = 1, Y_2 = 1 \mid X = 1) &= \frac{\Pr(Y_1 = 1, Y_2 = 1, X = 1)}{\Pr(X = 1)} && \text{(definition of conditional probability)} \\
&= \frac{0}{1/2} && \text{(two heads make } X = 2\text{, not 1)} \\
&= 0 && \text{(divide)}
\end{aligned}$$

but

$$
\begin{aligned}
\Pr(Y_1 = 1 \mid X = 1) &= \frac{\Pr(Y_1 = 1, X = 1)}{\Pr(X = 1)} && \text{(definition of conditional probability)} \\
&= \frac{\Pr(Y_1 = 1, Y_2 = 0)}{\Pr(X = 1)} && (Y_1 = 1 \text{ and } X = 1 \text{ means } Y_2 = 0) \\
&= \frac{1/4}{1/2} && \text{(substitute)} \\
&= \tfrac{1}{2} && \text{(divide)}
\end{aligned}
$$

and likewise $\Pr(Y_2 = 1 \mid X = 1) = 1/2$, so the product of the conditional probabilities is $1/4 \neq 0$: $Y_1$ and $Y_2$ are independent, but not conditionally independent given $X$.

**Proposition 0.1** (Neither independence nor conditional independence implies the other)**.** *There are random variables $Y_1$, $Y_2$, and $X$ such that $Y_1$ and $Y_2$ are conditionally independent given $X$ but not independent, and there are random variables $Y_1$, $Y_2$, and $X$ such that $Y_1$ and $Y_2$ are independent but not conditionally independent given $X$.*

Proof

*Proof.* Example 0.3 gives the first case, and Example 0.4 gives the second. □

**Example 0.5** (Wet roads, umbrellas, and rain)**.** Consider Brian Hutchinson's intuitive illustration of conditional independence [5]:
Let $w$ denote wet roads, $u$ denote people carrying umbrellas, and $r$ denote rain.

- **Marginally, wet roads and umbrellas are not independent ($w \not\perp u$).** If you look out a window and see people holding open umbrellas, that increases the probability that the pavement outside is wet.
- **Conditionally on rain, they are independent ($w \perp u \mid r$).** Once you know with certainty whether it is currently raining, learning whether pedestrians are carrying umbrellas tells you nothing more about the state of the roads:

$$
\Pr(w, u \mid r) = \Pr(w \mid r)\ \Pr(u \mid r), \qquad \Pr(w \mid u, r) = \Pr(w \mid r)
$$

**Proposition 0.2** (Conditional independence when each $Y_i$ depends only on its own $X_i$)**.** *Let $\underset{\sim}{X} = (X_1, \ldots, X_n)$ be a discrete random vector, and let $Y_1, \ldots, Y_n$ be conditionally independent given $\underset{\sim}{X}$. Suppose also that each $Y_i$ depends on $\underset{\sim}{X}$ only through $X_i$: for every $\underset{\sim}{x} = (x_1, \ldots, x_n)$ with $\Pr(\underset{\sim}{X} = \underset{\sim}{x}) > 0$ and every set of real numbers $A_i$,*

$$\Pr(Y_i \in A_i \mid \underset{\sim}{X} = \underset{\sim}{x}) = \Pr(Y_i \in A_i \mid X_i = x_i)$$

*Then, for every such* $\underset{\sim}{x}$ *and all sets of real numbers* $A_1, \ldots, A_n$*:*

$$\Pr(Y_1 \in A_1, \ldots, Y_n \in A_n \mid \underset{\sim}{X} = \underset{\sim}{x}) = \prod_{i=1}^{n} \Pr(Y_i \in A_i \mid X_i = x_i)$$

**Proof**

*Proof.* Each conditional probability given $X_i = x_i$ is defined, because $\Pr(X_i = x_i) \geq \Pr(\underset{\sim}{X} = \underset{\sim}{x}) > 0$. So:

$$\begin{aligned}
\Pr(Y_1 \in A_1, \ldots, Y_n \in A_n \mid \underset{\sim}{X} = \underset{\sim}{x}) &= \prod_{i=1}^{n} \Pr(Y_i \in A_i \mid \underset{\sim}{X} = \underset{\sim}{x}) && \text{(conditional independence given } \underset{\sim}{X}) \\
&= \prod_{i=1}^{n} \Pr(Y_i \in A_i \mid X_i = x_i) && \text{(each } Y_i \text{ depends on } \underset{\sim}{X} \text{ only through } X_i)
\end{aligned}$$

□

**Remark**

*Remark.* In regression, $\underset{\sim}{X}$ is usually the full set of covariates $(X_1, \ldots, X_n)$, and a model usually also assumes, as Proposition 0.2 does, that each $Y_i$ depends on the covariates only through its own $X_i$. That second assumption is a separate one, not part of conditional independence.

**Definition 0.3** (Identically distributed)**.** Random variables $X_1, \ldots, X_n$ are **identically distributed** if they all have the same CDF:

$$\forall i \in \{1, \ldots, n\}, \forall x \in \mathbb{R} : \Pr(X_i \leq x) = \Pr(X_1 \leq x)$$

**Theorem 0.3** (Identically distributed discrete random variables have the same PMF)**.** *Discrete random variables* $X_1, \ldots, X_n$ *are identically distributed if and only if they all have the same PMF: for every* $i \in \{1, \ldots, n\}$ *and every real number* $x$*,*

$$\mathbb{P}(X_i = x) = \mathbb{P}(X_1 = x)$$

Proof

*Proof.* Write $F_i(t) \stackrel{\text{def}}{=} \Pr(X_i \le t)$ for the CDF of $X_i$.

**Only if: the CDF determines the PMF through its jumps.** Choose an index $i$ and a value $x$ of $X_i$. Every number below $x$ lies in exactly one of the intervals $(-\infty, x-1]$, $(x-1, x-\frac{1}{2}]$, $(x-\frac{1}{2}, x-\frac{1}{3}]$, ..., so the event $\{X_i < x\}$ is the disjoint union of the events $B_1 \stackrel{\text{def}}{=} \{X_i \le x-1\}$ and $B_k \stackrel{\text{def}}{=} \{x - \frac{1}{k-1} < X_i \le x - \frac{1}{k}\}$ for $k \ge 2$, and for each $K$, the events $B_1, \ldots, B_K$ have union $\{X_i \le x - \frac{1}{K}\}$:

$$\begin{aligned}
\Pr(X_i < x) &= \sum_{k=1}^{\infty} \Pr(B_k) && \text{(countable additivity)} \\
&= \lim_{K\to\infty} \sum_{k=1}^{K} \Pr(B_k) && \text{(definition of an infinite series)} \\
&= \lim_{K\to\infty} \Pr(X_i \le x - \tfrac{1}{K}) && \text{(finite additivity)} \\
&= \lim_{K\to\infty} F_i(x - \tfrac{1}{K}) && \text{(definition of the CDF)}
\end{aligned}$$

The event $\{X_i \le x\}$ is the disjoint union of $\{X_i < x\}$ and $\{X_i = x\}$, so:

$$\begin{aligned}
\mathbb{P}(X_i = x) &= \Pr(X_i \le x) - \Pr(X_i < x) && \text{(finite additivity)} \\
&= F_i(x) - \Pr(X_i < x) && \text{(definition of the CDF)} \\
&= F_i(x) - \lim_{K\to\infty} F_i(x - \tfrac{1}{K}) && \text{(the display above)} \\
&= F_1(x) - \lim_{K\to\infty} F_1(x - \tfrac{1}{K}) && (F_i = F_1) \\
&= \mathbb{P}(X_1 = x) && \text{(the same two steps, for } X_1)
\end{aligned}$$

**If: the PMF determines the CDF through sums.** Let $S$ be the set of values $u$ with $\mathbb{P}(X_1 = u) > 0$. Since the PMFs are equal, $S$ is also the set of $u$ with $\mathbb{P}(X_i = u) > 0$, so $S \subseteq \mathcal{R}(X_i)$ and $S \subseteq \mathcal{R}(X_1)$. For every real $t$, the event $\{X_i \le t\}$ is the disjoint union of the events $\{X_i = u\}$ over the countably many $u \in \mathcal{R}(X_i)$ with $u \le t$, so:

$$\begin{aligned}
F_i(t) &= \sum_{u \in \mathcal{R}(X_i),\, u \le t} \mathbb{P}(X_i = u) && \text{(countable additivity)} \\
&= \sum_{u \in S,\, u \le t} \mathbb{P}(X_i = u) && \text{(drop the terms that are 0)} \\
&= \sum_{u \in S,\, u \le t} \mathbb{P}(X_1 = u) && \text{(the PMFs are equal)} \\
&= \sum_{u \in \mathcal{R}(X_1),\, u \le t} \mathbb{P}(X_1 = u) && \text{(restore the terms that are 0)} \\
&= F_1(t) && \text{(countable additivity)}
\end{aligned}$$

□

Remark

*Remark.* As with independence (Example 0.2), the PMF form would not work as a definition in general: every continuous random variable has $\Pr(X_i = x) = 0$ at every $x$.

**Example 0.6** (Identically distributed but not independent)**.** In Example 0.1, $X_1$ and $1 - X_1$ (the indicator of tails on the first flip) both take the values 0 and 1 with probability $1/2$ each, so they are identically distributed (Theorem 0.3). They are not independent: knowing one determines the other.

**Definition 0.4** (Conditionally identically distributed)**.** Random variables $Y_1, \ldots, Y_n$ are **conditionally identically distributed** given random variables $X_1, \ldots, X_n$ if the conditional CDF of each $Y_i$ given $X_i = x$ is one shared function $G(y \mid x)$ of $y$ and $x$: for every $i \in \{1, \ldots, n\}$, every $y$, and every $x$ with $\Pr(X_i = x) > 0$,

$$\Pr(Y_i \leq y \mid X_i = x) = G(y \mid x)$$

When $X_i$ is continuous, the condition applies at every $x$ with $\mathbb{p}(X_i = x) > 0$, with $\Pr(Y_i \leq y \mid X_i = x)$ computed from densities: the integral of $\mathbb{p}(X_i = x, Y_i = u)/\mathbb{p}(X_i = x)$ over $u \leq y$ (a sum over values $u \leq y$ when $Y_i$ is discrete).

**Example 0.7** (A shared regression model)**.** Suppose each $Y_i$ is binary, with $\Pr(Y_i = 1 \mid X_i = x) = \pi(x)$ for one function $\pi$ shared by every $i$ (for instance, $\pi(x) = x/(1 + x)$ for a dose $x \geq 0$). Then $Y_1, \ldots, Y_n$ are conditionally identically distributed given $X_1, \ldots, X_n$, with $G(y \mid x) = 1 - \pi(x)$ for $0 \leq y < 1$ (and 0 for $y < 0$, 1 for $y \geq 1$). Their marginal distributions can still differ: if participant 1 always receives dose 0 and participant 2 always receives dose 1, then $\Pr(Y_1 = 1) = 0$ but $\Pr(Y_2 = 1) = 1/2$.

**Definition 0.5** (Independent and identically distributed)**.** Random variables $X_1, \ldots, X_n$ are **independent and identically distributed** (shorthand: "$X_i$ iid") [6] if they are:

- statistically independent, and
- identically distributed.

Remark

*Remark.* The IID assumption is one of the most common assumptions in introductory statistics: it says a sample $X_1, \ldots, X_n$ can be treated as $n$ independent draws from a single shared distribution.

**Example 0.8** (Repeated die rolls)**.** The results of $n$ rolls of the same fair die are IID: the rolls are independent, and each is uniform on $\{1, \dots, 6\}$.

**Definition 0.6** (Conditionally independent and identically distributed)**.** Random variables $Y_1, \dots, Y_n$ are **conditionally independent and identically distributed** given $X_1, \dots, X_n$ (shorthand: "$Y_i \mid X_i$ ciid" or just "$Y_i \mid X_i$ iid") if:

- $Y_1, \dots, Y_n$ are conditionally independent given $(X_1, \dots, X_n)$,
- each $Y_i$ depends on $(X_1, \dots, X_n)$ only through $X_i$, and
- $Y_1, \dots, Y_n$ are conditionally identically distributed given $X_1, \dots, X_n$.

**Example 0.9** (The usual regression assumption)**.** In Example 0.7, if the $Y_i$ are also conditionally independent given all the doses, and each $Y_i$ depends on the doses only through its own $X_i$, then $Y_i \mid X_i$ ciid, and the joint conditional PMF is $\prod_{i=1}^{n} \pi(x_i)^{y_i}(1-\pi(x_i))^{1-y_i}$: one shared function evaluated at each $(x_i, y_i)$.

Video lecture

Hutchinson's Probability Refresher (27 min) covers independence [7]. The login for the video site is posted on Canvas.

Watch

5MinuteAI explains independence in five minutes, following the University of Washington's CSE 312 [8].

## References


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